

Lab 5: Circular Convolution

Instructions:

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
2. All the tasks in each lab have to be in one main MATLAB file with ".m" extension, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
3. Plots should include x label, y label, legend, and titles
4. File name: ID_Lab5.m/ID_Lab5.rar or zip **and** ID_Lab5.pdf
5. If the report/code is found to be copied, zero marks will be awarded [**Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.**]

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1. Develop a MATLAB function to construct a circulant matrix C as was discussed in the lecture, given an N-point sequence x(n).
 - a) Use the Toeplitz function to implement matrix C. Your subroutine function should have the following format:

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function [C] = circInt(x,N)
% Circulant Matrix from an N-point sequence
% [C] = circInt(x,N)
% C = circulant matrix of size NxN
% x = sequence of length <= N
% N = size of circulant matrix
```

b) A Circulant matrix is a special case of a Toeplitz matrix. [Observe and see if this statement is true] Verify your function on the sequence: $x_1(n) = \{1, 3, 5, 7, 9, -7, -5, -3, -1\}$

c) Using the circulant matrix developed in part a, evaluate the circular convolution of the sequences done in the lecture: $\{2, 1, 2, 1\}$ and $\{1, 2, 3, 4\}$

2. The following four sequences are given:

$$x_1(n) = \{1, 3, 2, -1\}; \quad x_2(n) = \{2, 1, 0, -1\}; \quad x_3(n) = x_1(n) * x_2(n); \quad x_4(n) = x_1(n) \circledast x_2(n)$$

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a) Determine and sketch $x_3(n)$.

b) Using $x_3(n)$ alone, determine and sketch $x_4(n)$. Do not directly compute $x_4(n)$.

You can take the help of the function developed in question 1.
